SFIA 9 – a framework for cybersecurity skills

The global skills and competency framework for the digital world

Cybersecurity strategy and architecture

Cybersecurity strategy and leadership

		1	2	3	4	5	6	7
Strategic planning	ITSP				4	5	6	7
Information security	SCTY		2	3	4	5	6	7
Demand management	DEMM				4	5	6	
Stakeholder relationship management	RLMT				4	5	6	7

Cybersecurity architecture

		1	2	3	4	5	6	7
Requirements definition and management	REQM		2	3	4	5	6	
Enterprise and business architecture	STPL					5	6	7
Solution architecture	ARCH				4	5	6	
Data management	DATM		2	3	4	5	6	

Cybersecurity governance, risk and compliance

		1	2	3	4	5	6	7
Governance	GOVN						6	7
Risk management	BURM		2	3	4	5	6	7
Audit	AUDT		2	3	4	5	6	7
Information and data compliance	PEDP				4	5	6	
Information management	IRMG			3	4	5	6	7
Information assurance	INAS		2	3	4	5	6	7
Quality management	QUMG		2	3	4	5	6	7
Measurement	MEAS		2	3	4	5	6	

Cybersecurity research, intelligence and advice

		1	2	3	4	5	6	7
Consultancy	CNSL				4	5	6	7
Specialist advice	TECH				4	5	6	
Formal research	RSCH		2	3	4	5	6	
Vulnerability research	VURE		2	3	4	5	6	
Threat intelligence	THIN		2	3	4	5	6	

Cybersecurity change programmes

Cybersecurity change programmes

		1	2	3	4	5	6	7
Programme management	PGMG						6	7
Project management	PRMG				4	5	6	7
Delivery management	DEMG			3	4	5	6	
Organisational change management	CIPM		2	3	4	5	6	
Organisational change enablement	OCEN				4	5	6	

Secure software and systems development

Secure development lifecycle management

		1	2	3	4	5	6	7
Systems development management	DLMG				4	5	6	7
Systems and software lifecycle engineering	SLEN			3	4	5	6	7

Secure design and architecture

		1	2	3	4	5	6	7
Systems design	DESN		2	3	4	5	6	
Infrastructure design	IFDN		2	3	4	5	6	
Software design	SWDN		2	3	4	5	6	
Network design	NTDS		2	3	4	5	6	
Hardware design	HWDE		2	3	4	5	6	

Secure development and implementation

		1	2	3	4	5	6	7
Programming/software development	PROG		2	3	4	5	6	
Systems integration and build	SINT		2	3	4	5	6	
Real-time/embedded systems development	RESD		2	3	4	5	6	

Security testing and validation

		1	2	3	4	5	6	7
Functional testing	TEST	1	2	3	4	5	6	
Non-functional testing	NFTS	1	2	3	4	5	6	
Penetration testing	PENT		2	3	4	5	6	

Cybersecurity capability development

Cybersecurity talent management

		1	2	3	4	5	6	7
Performance management	PEMT				4	5	6	
Employee experience	EEXP				4	5	6	
Professional development	PDSV				4	5	6	
Workforce planning	WFPL				4	5	6	
Resourcing	RESC		2	3	4	5	6	

Cybersecurity education and training

		1	2	3	4	5	6	7
Knowledge management	KNOW		2	3	4	5	6	7
Learning design and development	TMCR		2	3	4	5		
Learning and development management	ETMG		2	3	4	5	6	7
Learning delivery	ETDL		2	3	4	5		
Teaching	TEAC		2	3	4	5	6	7
Subject formation	SUBF				4	5	6	7
Certification scheme operation	CSOP		2	3	4	5	6	

Cybersecurity operations and resilience

Secure infrastructure management

		1	2	3	4	5	6	7
Technology service management	ITMG					5	6	7
Infrastructure operations	ITOP	1	2	3	4	5		
Network support	NTAS	1	2	3	4	5		
Database administration	DBAD		2	3	4	5		
Release management	RELM		2	3	4	5	6	
Deployment	DEPL		2	3	4	5	6	
Storage management	STMG		2	3	4	5	6	
System software administration	SYSP		2	3	4	5		

Cybersecurity resilience

		1	2	3	4	5	6	7
Security operations	SCAD	1	2	3	4	5	6	
Identity and access management	IAMT	1	2	3	4	5	6	
Continuity management	COPL		2	3	4	5	6	
Incident management	USUP	1	2	3	4	5	6	
Problem management	PBMG		2	3	4	5		
Change control	CHMG		2	3	4	5	6	
Asset management	ASMG		2	3	4	5	6	
Vulnerability assessment	VUAS		2	3	4	5		
Digital forensics	DGFS		2	3	4	5	6	
Cybercrime investigation	CRIM		2	3	4	5	6	
Offensive cyber operations	OCOP		2	3	4	5	6	

Cybersecurity data and analytics

		1	2	3	4	5	6	7
Data analytics	DAAN		2	3	4	5	6	7
Data science	DATS		2	3	4	5	6	
Business intelligence	BINT		2	3	4	5		
Data engineering	DENG		2	3	4	5	6	

Supply chain

Secure supply chain

		1	2	3	4	5	6	7
Sourcing	SORC		2	3	4	5	6	7
Supplier management	SUPP		2	3	4	5	6	7
Contract management	ITCM		2	3	4	5	6	7

Generic attributes

Attributes

		1	2	3	4	5	6	7
Autonomy	AUTO	1	2	3	4	5	6	7
Complexity	COMP	1	2	3	4	5	6	7
Influence	INFL	1	2	3	4	5	6	7
Knowledge	KNGE	1	2	3	4	5	6	7

Business skills/Behavioural factors

		1	2	3	4	5	6	7
Collaboration	COLL	1	2	3	4	5	6	7
Communication	COMM	1	2	3	4	5	6	7
Improvement mindset	IMPM	1	2	3	4	5	6	7
Creativity	CRTY	1	2	3	4	5	6	7
Decision-making	DECM	1	2	3	4	5	6	7
Digital mindset	DIGI	1	2	3	4	5	6	7
Leadership	LEAD	1	2	3	4	5	6	7
Learning and development	LADV	1	2	3	4	5	6	7
Planning	PLAN	1	2	3	4	5	6	7
Problem-solving	PROB	1	2	3	4	5	6	7
Adaptability	ADAP	1	2	3	4	5	6	7
Security, privacy and ethics	SCPE	1	2	3	4	5	6	7

Security specialists need specialised skills in cybersecurity and also a range of other skills which are re-useable in the wider organisational context.

For professionals outside of the security domain, a foundational understanding of security principles is necessary, complementing their role-specific knowledge. They need the skills for their own specialism and the know-how to apply secure working practices within their roles. E.g. for risk management, software engineering, networks, technology infrastructure.

If you can't find a skill you are looking for, try the full SFIA framework.

SFIA provides a structured and consistent approach to defining cybersecurity skills.
Each skill is clearly described, supplemented by guidance notes, and detailed level-by-level practice descriptions that align with the framework's 7 levels of responsibility. This uniform structure ensures ease of navigation and understanding, seamlessly integrating professional skills with behavioural factors to outline comprehensive role expectations. The consistent detail across all levels ensures robustness, allowing for precise skills and competency assessment.
The clarity in describing the nuances of cybersecurity roles at every responsibility level makes it invaluable for developing and benchmarking cybersecurity capabilities within an organisation.

	SFIA Level 1 Follow	SFIA Level 2 Assist	SFIA Level 3 Apply	SFIA Level 4 Enable	SFIA Level 5 Ensure, advise	SFIA Level 6 Initiate, influence	SFIA Level 7 Set strategy, inspire, mobilise
SFIA Levels of responsibility	Performs routine tasks under close supervision, follows instructions, and requires guidance to complete their work. Learns and applies basic skills and knowledge.	Provides assistance to others, works under routine supervision, and uses their discretion to address routine problems. Actively learns through training and on-the-job experiences.	Performs varied tasks, sometimes complex and non-routine, using standard methods and procedures. Works under general direction, exercises discretion, and manages own work within deadlines. Proactively enhances skills and impact in the workplace.	Performs diverse complex activities, supports and guides others, delegates tasks when appropriate, works autonomously under general direction, and contributes expertise to deliver team objectives.	Provides authoritative guidance in their field and works under broad direction. Accountable for delivering significant work outcomes, from analysis through execution to evaluation.	Influences the organisation significantly, makes high-level decisions, shapes policies, demonstrates thought leadership, fosters collaboration, and accepts accountability for strategic initiatives and outcomes.	Operates at the highest organisational level, determines overall organisational vision and strategy, and assumes accountability for overall success.

SFIA's attributes of Autonomy, Influence and Complexity are the key to determining level of impact, responsibility and accountability. Click the SFIA level to find the details.

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